

KG RAG ASSIGNMENT

Hospital Q&A System with Knowledge Graph RAG

TASKS

Take the “hospital visits” pdf and “disease treatments” pdfs and:

1) Take it through the Graph **indexing** flow

A) Upload the PDFs to Neo4j Graph Builder to directly create graph from pdf, and

B) Read the pdfs using a pdf python library such as PyMupdf, clean the text and send it to LLM to create the nodes and relationships

Compare the 2 approaches and finalize on the one which is doing better.

2) Implement app in Python that will :

a) take a user question as input

b) Use LLM to generate cypher query based on the question

Hint: Pass the llm schema to the LLM in context

c) Send the cypher query to neo4j and obtain the results

d) send the results and prompt to LLM to generate final response.

Demonstrate with few questions such as:

1) What are the treatments for <disease>

2) Which doctors did patient X visit

3) Which diseases does drug Y treat

(at least 5)

Submit: screenshots of output and neo4j schema graph visualization